# KASHISH UPENDRARAJ SINGH

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## EDUCATION

**The University of Texas at Dallas, Richardson, TX** May 2027
*MS, Information Technology & Management* **GPA 3.89**

**University of Mumbai, SIES Graduate School of Technology, Navi Mumbai, India** July 2023
*Bachelor of Engineering, Information Technology* **GPA 3.8**

**Coursework:** Database Foundations, Applied Machine Learning, Big Data, Business Analytics, Advanced Statistics for Data Science, Spreadsheet Modeling

## PROFESSIONAL EXPERIENCE

**Federal Reserve Bank of Dallas** June 2026 – Present
*Software Engineer Intern*

- Engineered a rerunnable **Python** pipeline scoring technology exposure across 21,155 SEC EDGAR 10K filings (1,971 U.S. banks, 1994–2025), cutting manual data extraction time by **90%**.
- Explored and integrated five FFIEC datasets into a 2.5M observation branch panel in **R**, documenting methods and presenting findings to research economists.
- Automated a weekly AI news digest that keeps technical and non technical leadership current on the AI landscape.

**HDFC Bank, Mumbai, India** Aug 2023 – Aug 2025
*Software Engineer*

- Gathered business and functional requirements from operations stakeholders, then built and deployed an ensemble ML anomaly detection model across NetBanking and MobileBanking, reaching **95%** accuracy.
- Built real time Oracle GoldenGate lag monitoring in Cisco AppDynamics, reducing incident detection and resolution time (MTTD/MTTR) by **40%**.
- Automated transaction reporting on Microsoft Power Platform (Power BI, Power Automate), cutting a 20 minute daily job to under 10 seconds, about **87 hours** saved per year.
- Automated SIT testing with Pytest within the SDLC, cutting runtime **92%** and documenting test plans and processes.
- Delivered knowledge transfer sessions to new teammates and maintained process documentation for the team.

## TECHNICAL SKILLS

**Programming Languages:** Python, R, SQL, C, MATLAB, Shell Scripting
**Databases & Big Data:** MySQL, Oracle, MongoDB, Neo4j, ChromaDB, Apache Spark, Apache Kafka, Hadoop, DataBricks
**Cloud & Platforms:** Microsoft Azure, AWS, Docker, CI/CD, Git, Linux
**Data Visualization & BI:** Power BI, Power Automate (Power Platform), Tableau, Advanced Excel
**Framework & Tools:** Pandas, Scikit-learn, dplyr, statsmodels, Stata, Pytest, Selenium
**AI/ML:** Generative AI, LLMs, Prompt Engineering, RAG, LangChain, Embeddings, NLP, Time Series
**AI Tools:** Microsoft Copilot, GitHub Copilot, ChatGPT, Claude
**Methodologies:** Agile, SDLC, Requirements Gathering, Process Documentation, Knowledge Transfer

## PROJECTS

**Texas Grid Demand Forecasting** *(Python, XGBoost, Scikit-learn, SHAP)*

- Blended XGBoost with neural networks to forecast Texas power demand 24 hours ahead at **2.13%** error, **48%** better than baseline.
- Gives grid operators a reliable day ahead view of demand to prepare for heat waves and peak load.

**MedGraph AI: Medical Assistant** *(Python, Neo4j, ChromaDB, Streamlit)*

- Prototyped a generative AI proof of concept using prompt engineering and RAG to map symptoms to likely conditions.
- Demoed a Streamlit app that flags emergencies and creates doctor ready reports for patients.

**Incident Prediction System** *(Python, Machine Learning, AWS, AppDynamics, SQL)*

- Tuned classification models to flag banking incidents pre impact, presenting results to IT leadership.
- Deployed on AWS real time pipelines for automated early warning alerts in production.

**Catalyst: Student Performance Tracker** *(Node.js, EJS, MongoDB, Python)*

- Built a full stack web application tracking academic records for students, teachers, and parents.
- Added a Machine Learning career prediction feature (**90% accuracy**) recommending academic paths to students.